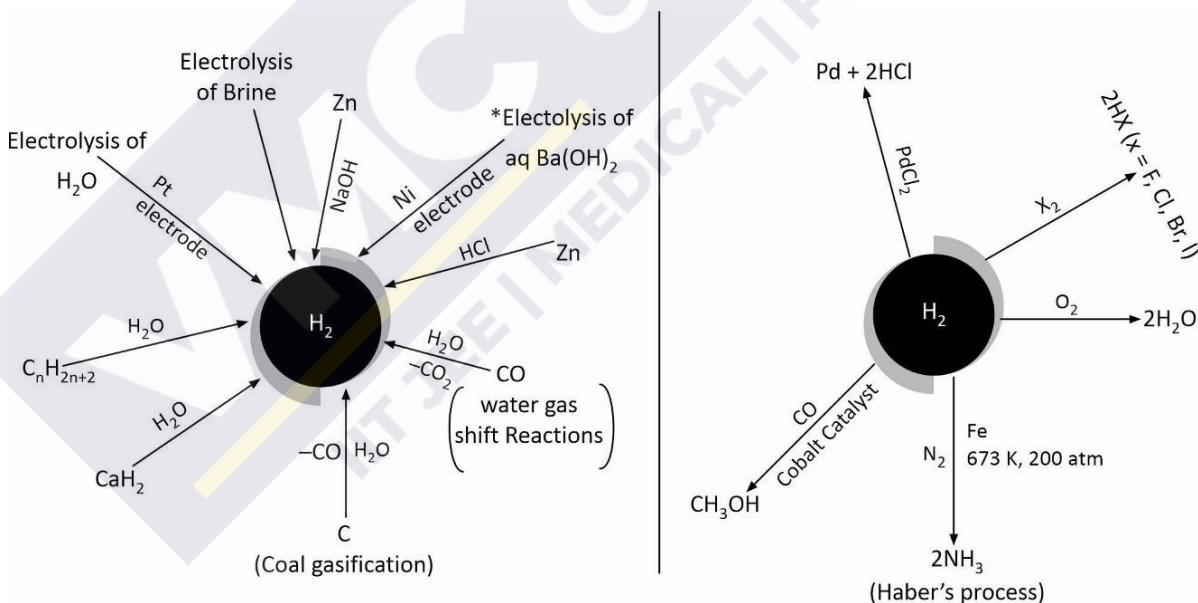


Inorganic Memory Chart

1. Dihydrogen, H_2

Atomic and Physical Properties of Hydrogen

Property	Hydrogen	Deuterium	Tritium
Relative abundance (%)	99.985	0.0156	10^{-15}
Relative atomic mass ($g\ mol^{-1}$)	1.008	2.014	3.016
Melting point / K	13.96	18.73	20.62
Boiling point / K	20.39	23.67	25.0
Density / $g\ L^{-1}$	0.09	0.18	0.27
Enthalpy of fusion / $kJ\ mol^{-1}$	0.117	0.197	-
Enthalpy of vaporization / $kJ\ mol^{-1}$	0.904	1.226	-
Enthalpy of bond dissociation / $kJ\ mol^{-1}$ at 298.2K	435.88	443.35	-
Internuclear distance / pm	74.14	74.14	-
Ionization enthalpy $kJ\ mol^{-1}$	1312	-	-
Electron gain enthalpy / $kJ\ mol^{-1}$	-73	-	-
Covalent radius / pm	37	-	-
Ionic radius (H^-) / pm	208		



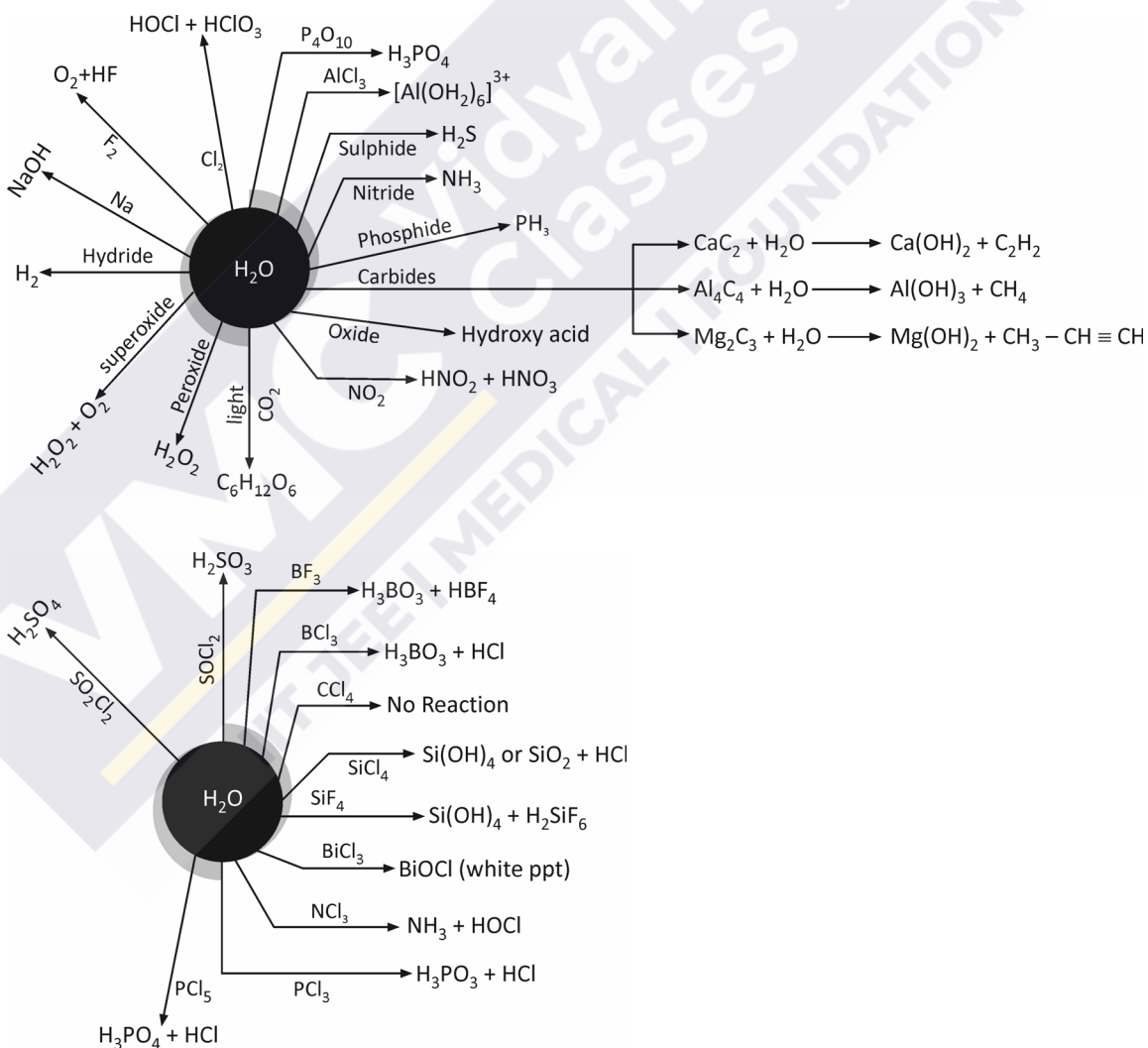
* **Very pure hydrogen is obtained**

- ☐ Colourless, tasteless, odourless gas
- ☐ low solubility in liquid solvents
- ☐ Inert at room temperature due to high H-H bond enthalpy.

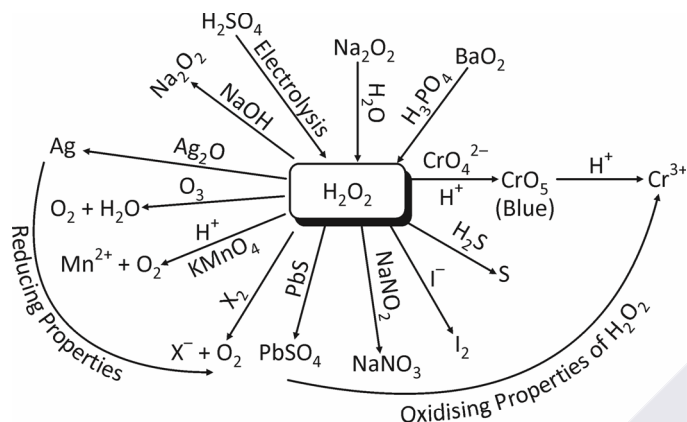
2. Water (H₂O)

 Physical Properties of H₂O and D₂O

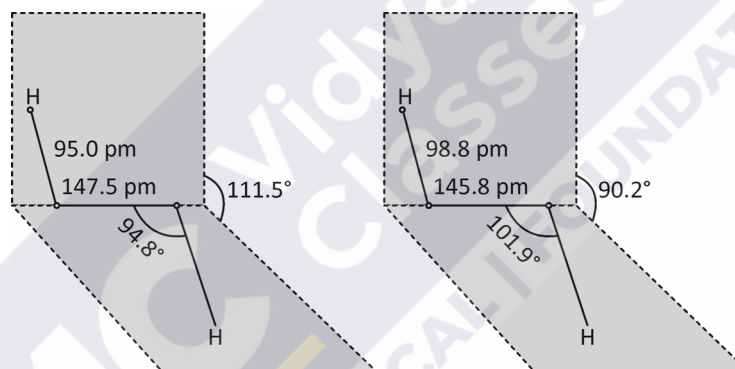
Property	H ₂ O	D ₂ O
Molecular mass (g mol ⁻¹)	18.0151	20.0276
Melting point / K	273.0	276.8
Boiling point / K	373.0	374.4
Enthalpy of formation / kJ mol ⁻¹	-285.9	-294.6
Enthalpy of vaporization (373K) / kJ mol ⁻¹	40.66	41.61
Enthalpy of fusion / kJ mol ⁻¹	6.01	-
Temp of max. density / K	276.98	284.2
Density (298K)/g cm ⁻³	1.0000	1.1059
Viscosity / centipoise	0.8903	1.107
Dielectric constant / C ² /N.m ²	78.39	78.06
Electrical conductivity (293 K /ohm ⁻¹ cm ⁻¹)	5.7 × 10 ⁻⁸	-



3. HYDROGEN PEROXIDE (H_2O_2)



Melting point / K	272.4	Density (liquid at 298 K) / g cm^{-3}	1.44
Boiling point (extrapolated)/K	423	Viscosity (290K) / centipoise	1.25
Vapour pressure (298 K) / mmHg	1.9	Dielectric constant (298 K) / $\text{C}^2/\text{N m}^2$	70.7
Density (solid at 268.5K)/ g cm^{-3}	1.64	Electrical conductivity (298K) / $\Omega^{-1}\text{cm}^{-1}$	5.1×10^{-8}



(a) Gas phase

(b) Solid phase

(a) H_2O_2 structure in gas phase, dihedral angle 111.5°

(b) H_2O_2 structure in solid phase at 110K, dihedral angle is 90.2°

1. Periodic Properties of IA (1) and II A(2) group elements

A : Atomic number

B : Effective nuclear charge ($Z^* = Z - \sigma$)

Where σ is screening constant

C : Number of shells

D : Atomic radius

E : Ionisation energy

F : Hydration and hydration energy

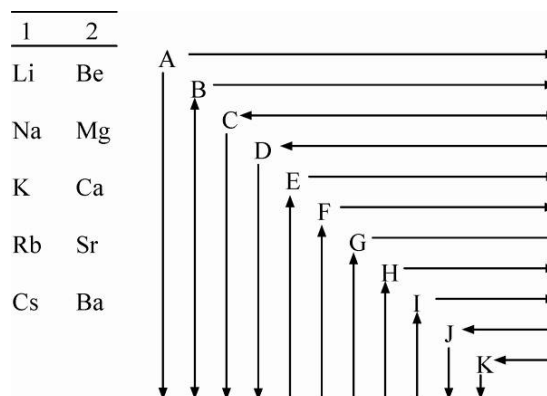
G : Size of hydrated ion

H : Polarising power (based on Fajan's rule)

I : Lattice energy of chloride

J : Reactivity

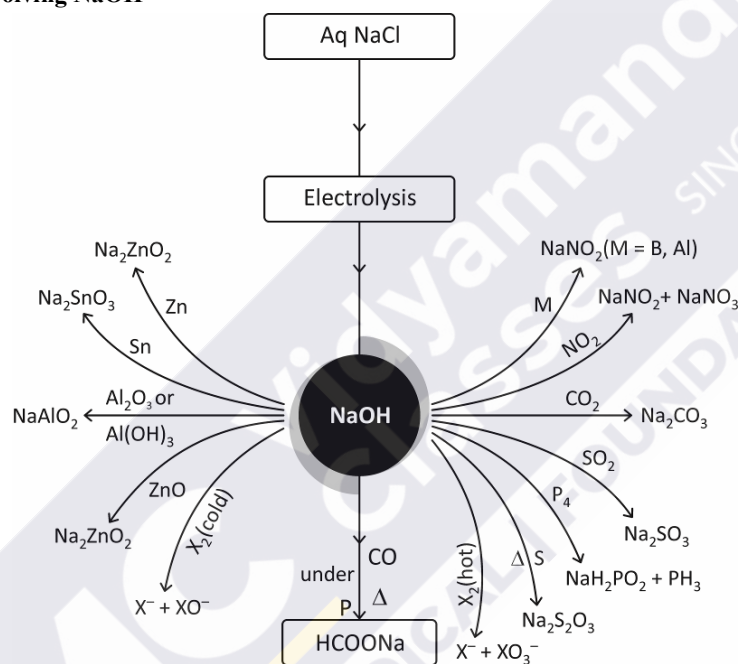
K : Ionic conductance



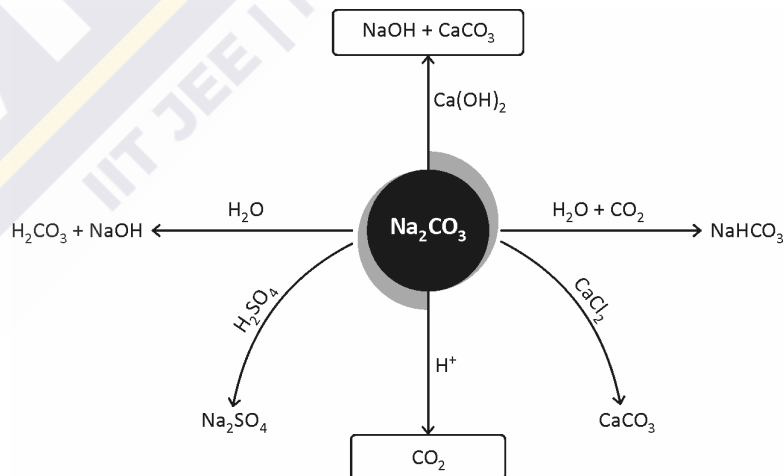
2. Gradation in the Properties of Alkali Metals :

Atomic radii	Li	M.P. and B.P.	max
Atomic volume	Na	Hardness	↑
Density	K	Ionisation energy	↑
Reactivity	Rb	Conductivity	↑
Reducing power	Cs	Electronegativity	↑
Electropositivity		(Solubility of salts having large anions)	↑
Anion stabilisation (Solubility of salts having small anions)			
	max		

3. Reactions Involving NaOH



4. Reactions Involving Na_2CO_3



5. Distinction between IIA(2) and IIB (12) elements :

Properties	IIA (Be, Mg, Ca, Sr, Ba, Ra)	II B (Zn, Cd, Hg)
1. Electronic Configuration	[Inert gas] ns^2	[Inert gas] $(n-1)d^{10} ns^2$
2. Block	s-block	d-block
3. Oxidation state	+2	+2, mercury also forms dimeric Hg_2^{2+}
4. Nature of oxide	BeO is amphoteric, other oxides are basic Basic nature is in order. $MgO < CaO < SrO < BaO < RaO$	ZnO is amphoteric, CdO and HgO are basic
5. Nature of halides	Electron-deficient BeX_2 covalent, others (MX_2) are Ionic : $MgCl_2 < CaCl_2 < SrCl_2 < BaCl_2$	$ZnCl_2$, $CdCl_2$ are ionic but less than IIA; $HgCl_2$ is covalent.
6. Nature of sulphates	Less soluble in water and solubility decreases down the group $BeSO_4 > MgSO_4 > CaSO_4 > SrSO_4 > BaSO_4$	more soluble than IIA Solubility of hydroxides decreases as we move down the group.
7. Nature of hydroxides	Solubility of hydroxides increases as we move down the group	ZnS , CdS , HgS insoluble and precipitate in salt analysis.
8. Nature of sulphides	Soluble	Decreases as we move down the group $Zn > Cd > Hg$
9. Reactivity	Increases as we move down the group $Be < Mg < Ca < Sr < Ba$	

6. Trends in properties of p - Block Elements in the direction of arrow

x, y, t ↓	p, q, r →					
	B	C	N	O	F	Ne
	Al	Si	P	S	Cl	Ar
	Ga	Ge	As	Se	Br	Kr
	In	Sn	Sb	Te	I	Xe
	Tl	Pb	Bi	Po	At	Rn
← x, y, z (upto group 14)						p, z ↑ q, r s (except for N_2, O_2, F_2)

where represents

x : covalent / van der Waals' radius

z : heat of sublimation

q : ionisation energy ($N > O, P > S$)

s : stability of higher oxidation state within a group

y : metallic character

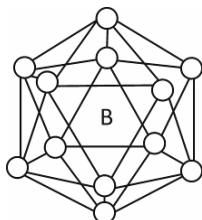
p : electronegativity

r : oxidising power

t : stability of lower oxidation state within a group

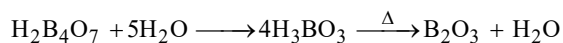
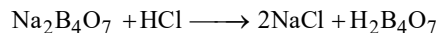
BORON :

- B exist in many allotropic forms
- All allotropes have B_{12} icosahedra units made up of polyhedron having 20 faces and 12 corners.
- Simplest form : α - Rhombohedral

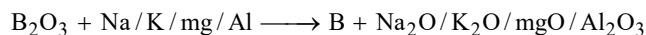


Isolation of "B"

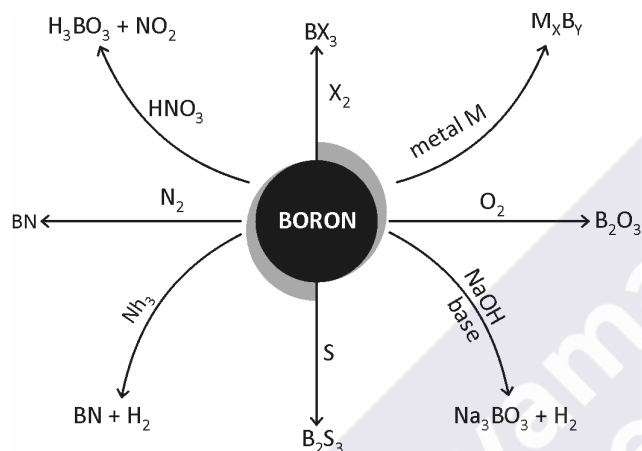
(i) Preparation of B_2O_3 from Borax / Colemanite



(ii) Reduction of B_2O_3

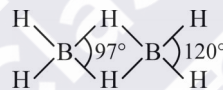


7. Reactions Involving Boron :

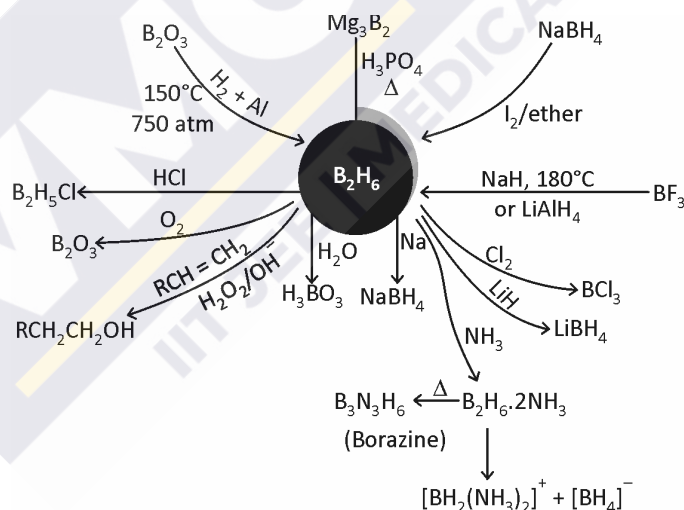


Diborane

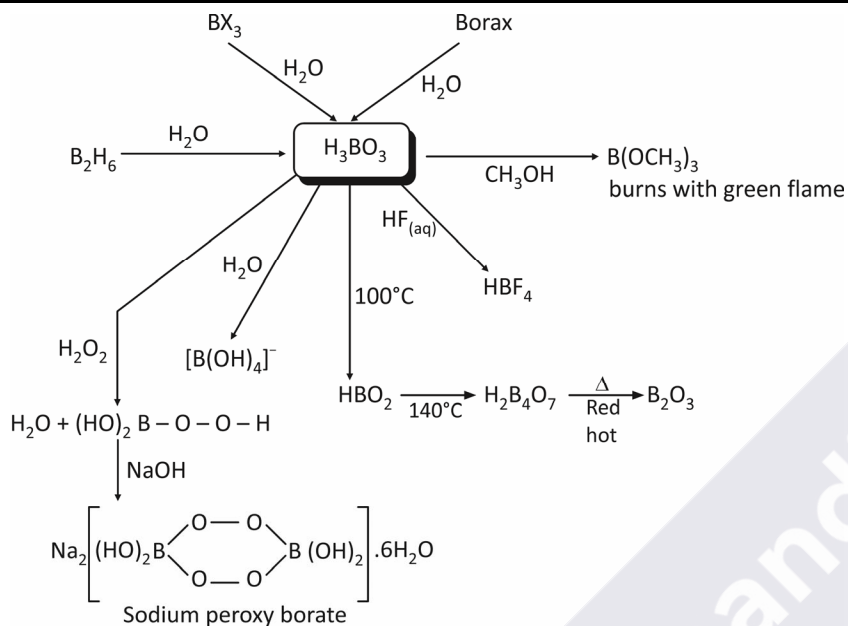
- $3c - 2e^-$ B – H – B bond (Banana bond) : 2
- Non – planar Molecule
- No. of atoms in 1 plane : 6
- e^- deficient Molecule



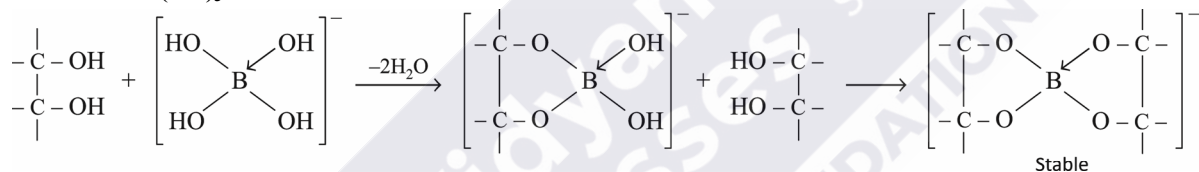
8. Reactions involving Diborane



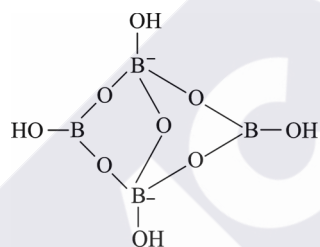
9. **Ortho Boric acid** : weak monobasic acid
 (H_3BO_3) : lewis acid
Structure :



Titration of $B(OH)_3$ with $NaOH$:

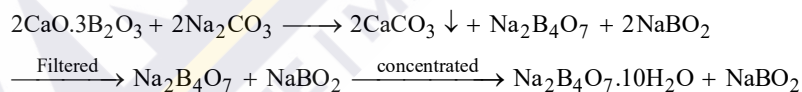


10. Borax :

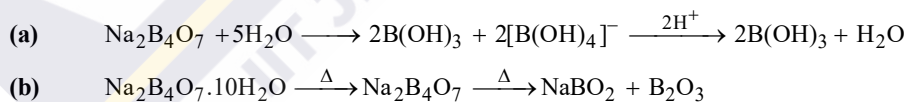


There are two tetrahedral and two triangular units in $[B_4O_5(OH)_4]^{2-}$.

Preparation :

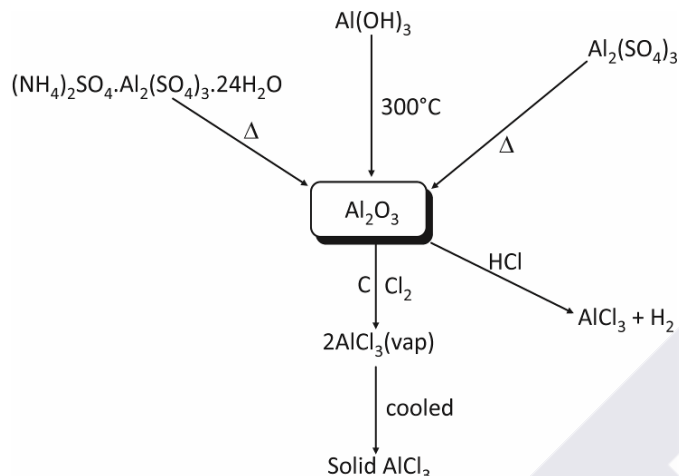


Reaction :



11. Alumina : (Al_2O_3)- White crystalline Powder – amphoteric

- Insoluble in water - α form is known as corundum

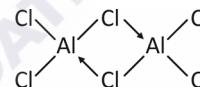


12. AlCl_3 :

- Colourless crystalline solid ; - Soluble in water

Structure :

- In Gas phase : (C.N = 4) : Exist as dimer
 - Both Al are sp^3 hybridized ; - 2 ($3\text{c} - 4\text{e}^-$) Al – Cl – Al bond
 - 4 ($2\text{c} - 2\text{e}^-$) Al – Cl bond
 - Bridge bond is longer than terminal bond
- In aq. Phase, it exist as $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ (C.N = 6)
- In solid state, at low temp, AlCl_3 exist as closed packed lattice of Cl^- with Al^{3+} Occupying Octahedral lattice (C.N = 6)



13. Alums : $\text{M}_2\text{SO}_4 \cdot \text{M}'_2(\text{SO}_4)_2 \cdot 24\text{H}_2\text{O}$

$\text{M} = \text{Na}^+, \text{K}^+, \text{Rb}^+, \text{Cs}^+, \text{As}^+, \text{NH}_4^+, \text{Ti}^+$;

$\text{M}' = \text{Al}^{3+}, \text{Cr}^{3+}, \text{Fe}^{3+}, \text{Mn}^{3+}, \text{Co}^{3+}$

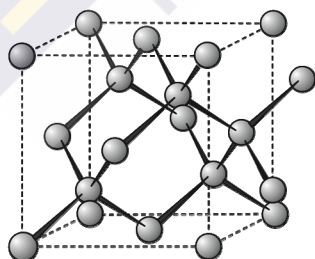
Preparation $\text{Al}_2\text{O}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{Al}_2(\text{SO}_4)_3 \xrightarrow{\text{K}_2\text{SO}_4(\text{aq})} \text{Crystallise (Double salt) i.e Alum}$

CARBON :

Allotropes : Graphite, Diamond, Fullerene

Diamond

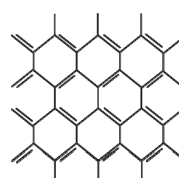
- ❑ Extremely unreactive, colourless, non conductor and shows brilliance due to total internal reflection.
- ❑ The structure comprises of sp^3 hybrid C atoms forming σ -bonds and arranged tetrahedrally.



The crystal structure of diamond.

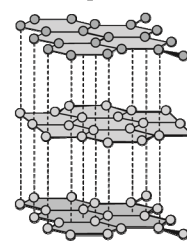
Graphite

- ❑ Quite reactive and soft.
- ❑ It has a layer structure with layers held together by weak van der waal's forces. Each sheet consists of sp^2 hybridised carbon atoms which are covalently bonded to three carbon atoms by σ -bonds. The fourth e^- is in unhybridised 'p' orbital and forms a partial ' π ' bond



(a)

(a) The structure of a graphite sheet.



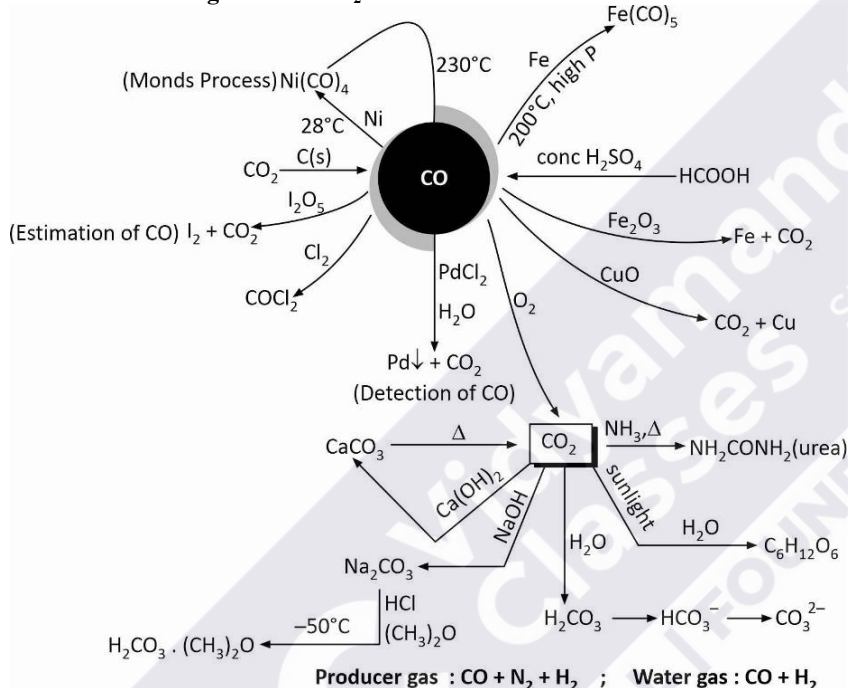
(b)

(b) Structure of a-graphite

It is used as a lubricant due to slippery layers and electricity is conducted only along sheets.

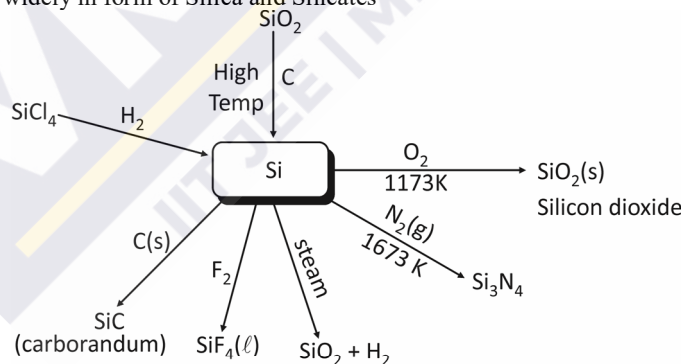
Physical properties	Graphite	Diamond
Density (g cm^{-3})	2.26	3.51
Enthalpy of combustion to yield CO_2 (kJ/mol)	-393.3	-395.5
Melting point ($^{\circ}\text{C}$)	2820	3730
Boiling point ($^{\circ}\text{C}$)		4830
Conductivity (electrical)	Fairly good	Non-conductor
Conductivity (thermal)	good	Max-conductivity

14. Reactions involving CO and CO_2

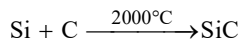
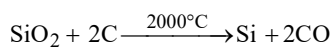


15. Silicon :

- 2nd most abundant element on Earth crust
- does not occurs in free state
- Occur widely in form of Silica and Silicates



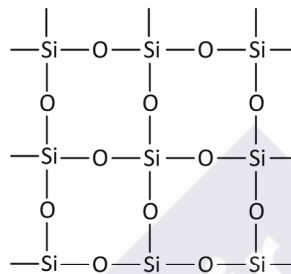
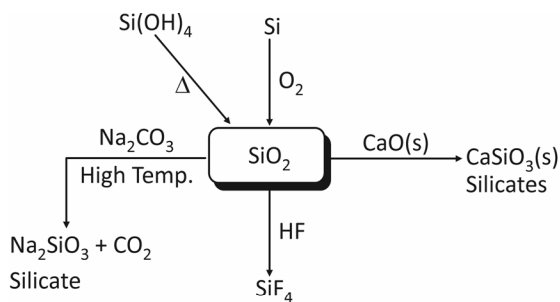
Silicon Carbides (SiC)



- Very hard
- Inert

16. Silica (SiO₂)

- Three dimensional network structure
- Si is sp³ hybridised



17. Silicates (SiO₄⁴⁻ tetrahedral unit)

- exist as discrete unit or may Polymerise into larger unit by sharing Oxygen

Classification of Silicate

	Gen Formula	No. of Shared 'O'	Example
Orthosilicate	SiO ₄ ⁴⁻	0	Mg ₂ SiO ₄
Pyrosilicate	[Si ₂ O ₇] ⁶⁻	1	SC ₂ Si ₂ O ₇
Cyclic Silicate	(SiO ₃) _n ²ⁿ⁻	2	LiAl(SiO ₃) ₂
Chain Silicate	(SiO ₃) _n ²ⁿ⁻	2	
Double chain silicate (Amphiboles)	(Si ₄ O ₁₁) _n ⁶ⁿ⁻	2.5	Ca ₂ Mg ₅ (Si ₄ O ₁₁) ₂ (OH) ₂
2D – Sheet Silicate	(Si ₂ O ₅) _n ²ⁿ⁻	3	Al ₂ (OH) ₄ (Si ₂ O ₅)
3D – Sheet Silicate	SiO ₂	4	Quartz, Zeolites

Note : General formula : $\left[\text{SiO}_{4-\frac{x}{2}} \right]^{4-x}$
(Hence x is no of shared oxygen)

18. Allotropes of Tin

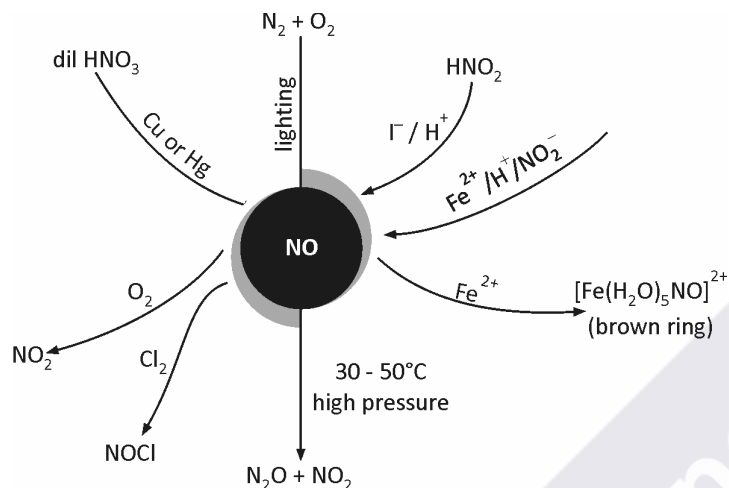
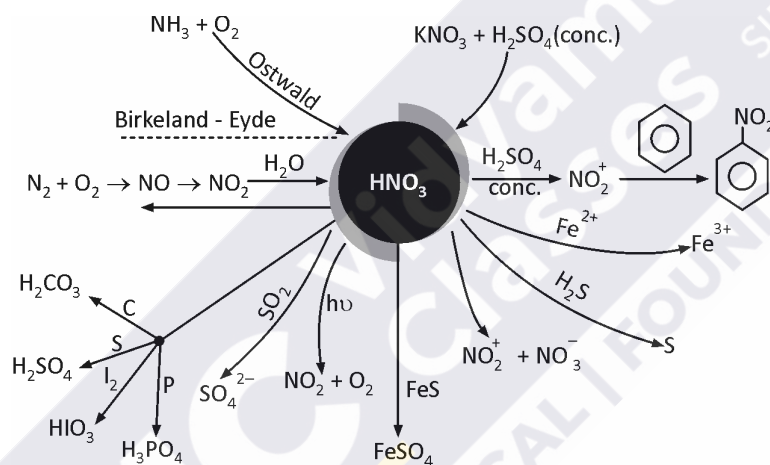
- ♦ Alpha or Grey Tin[Sn]
Tin exists in this form at low temperatures, below 13.2°C. It has a cubic crystal structure similar to silicon and germanium.
- ♦ Beta or White Tin[Sn]
Tin exists in this form when heated above 13.2°C. It is metallic and has a tetragonal structure.
It slowly changes back to the grey form when cooled, which is called the "tin pest" or "tin disease".



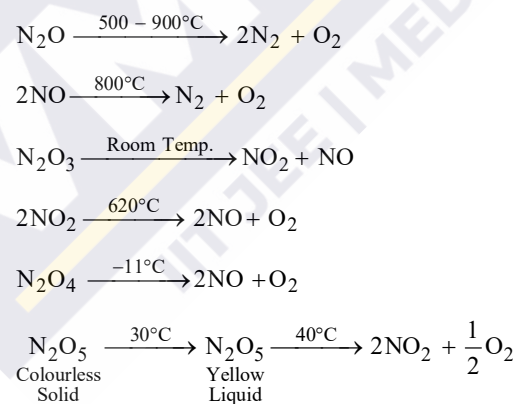
21. Reaction involving N_2O (Laughing Gas)



22. Reactions involving NO (Nitric Oxide)


 23. Reactions Involving HNO_3 (Nitric Acid)


Decomposition Behaviour



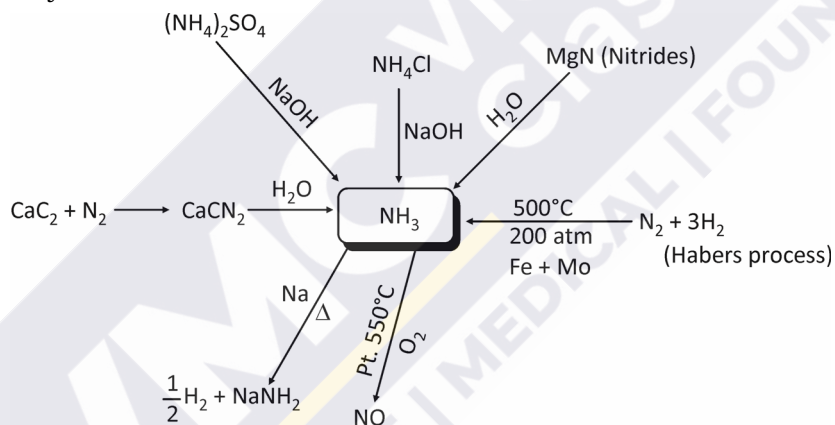
Reaction with H₂O & NaOH

	H ₂ O	NaOH
N ₂ O	Neutral Solution	-
NO	Neutral Solution	-
N ₂ O ₃	2HNO ₂	2NaNO ₂
NO ₂	HNO ₂ + HNO ₃	NaNO ₂ + NaNO ₃
N ₂ O ₅	2HNO ₃	2NaNO ₃

Reaction of Nitric acid with Metal

S. No.	Conc of HNO ₃	Metal	Product
1.	Very dilute HNO ₃ (6%)	Mg, Mn Fe, Zn, Sn	H ₂ + Metal Nitrate NH ₄ NO ₃ + Metal Nitrate + H ₂ O
2.	Dilute HNO ₃ (~ 20%)	Cu, Ag, Hg Fe, Zn Sn	NO + Metal Nitrate + H ₂ O N ₂ O + Metal Nitrate + H ₂ O NH ₄ NO ₃ + Sn(NO ₃) ₂
3.	Conc. HNO ₃ (~ 10%)	All Metals Sn	NO ₂ + Metal Nitrates + H ₂ O NO ₂ + H ₂ SnO ₃

24. NH₃



25. Phosphorous

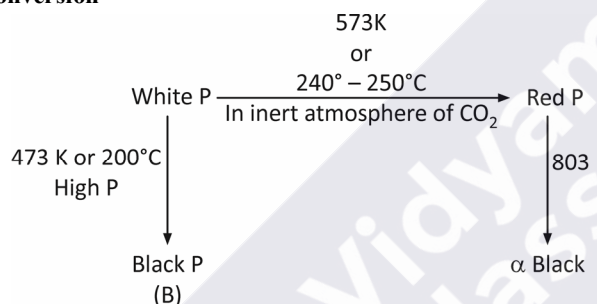
- Allotropes

- (a) White or yellow Phosphorous (P₄) → (tetrahedral structure)
- (b) Red Phosphorous → (linear chain)
- (c) Black Phosphorous → (Sheet structure)

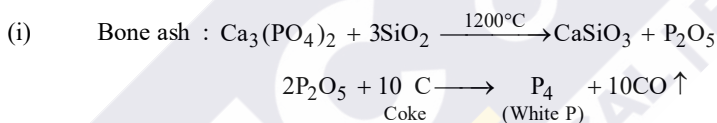
Reactivity : Black < Red < White

White phosphorus	Red phosphorus	Black phosphorus
Discrete tetrahedral P ₄ molecules	Polymeric structure consisting of chains of P ₄ units linked together	Exists in two forms α-black P and β-black P
Very reactive	Less reactive than white P	Very less reactive
Glow in dark	Does not glow in dark	
Translucent waxy solid	Has an iron grey luster	Has an opaque monoclinic or rhombohedral crystals
Soluble in CS ₂ but insoluble in water	Insoluble in water as well as CS ₂	
It has low ignition temperature, therefore, kept under water		

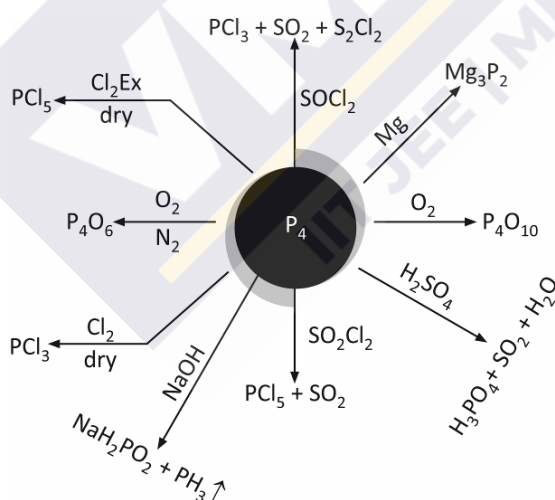
Inter-conversion



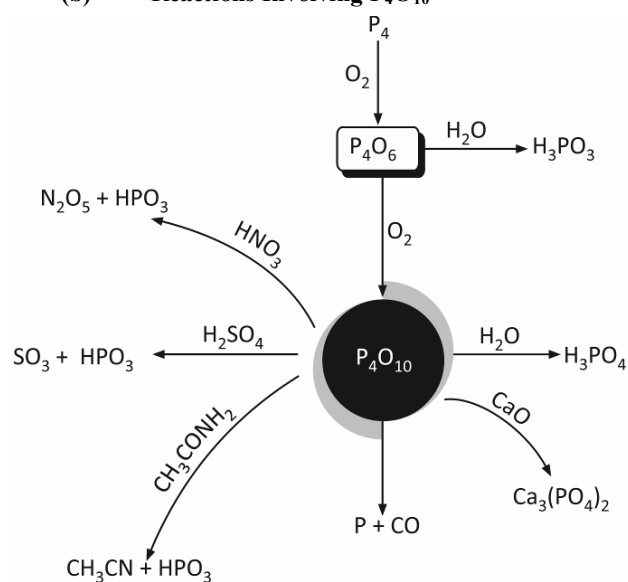
Preparation (P₄)



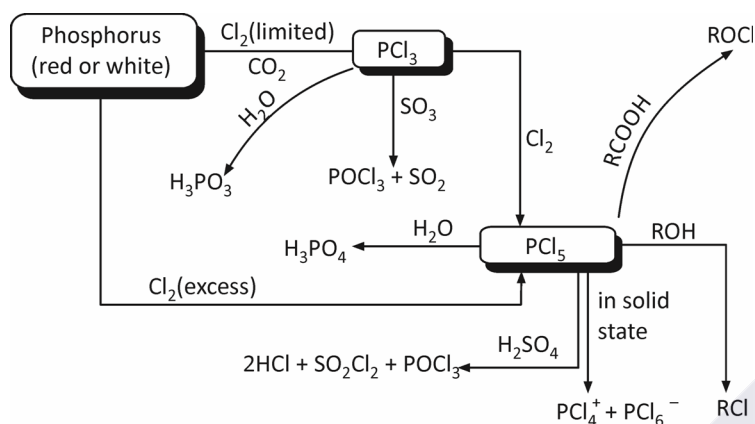
26. (a) Reaction of P₄



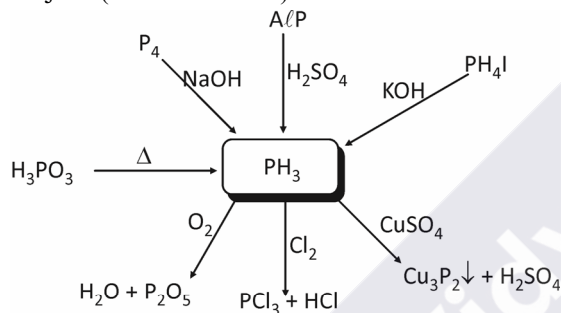
(b) Reactions Involving P₄O₁₀



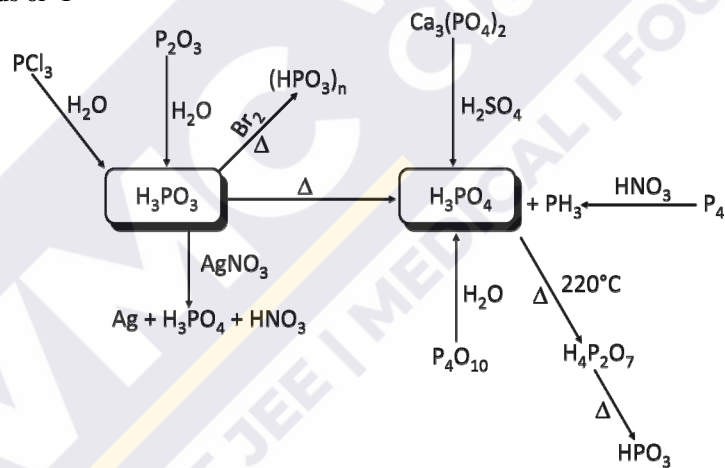
27. Reactions Involving Phosphorus chlorides.



28. PH_3 (Rotten fish smell)

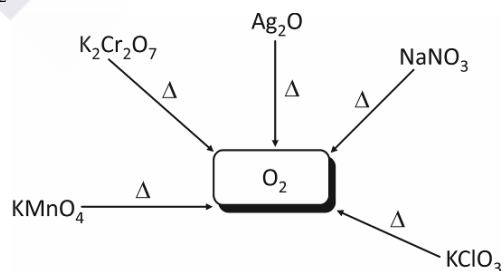


29. Oxyacids of 'P'



30. Oxygen

Allotropes : O_2 and O_3
 Liquid O_2 : Pale blue
 Solid O_2 : Blue



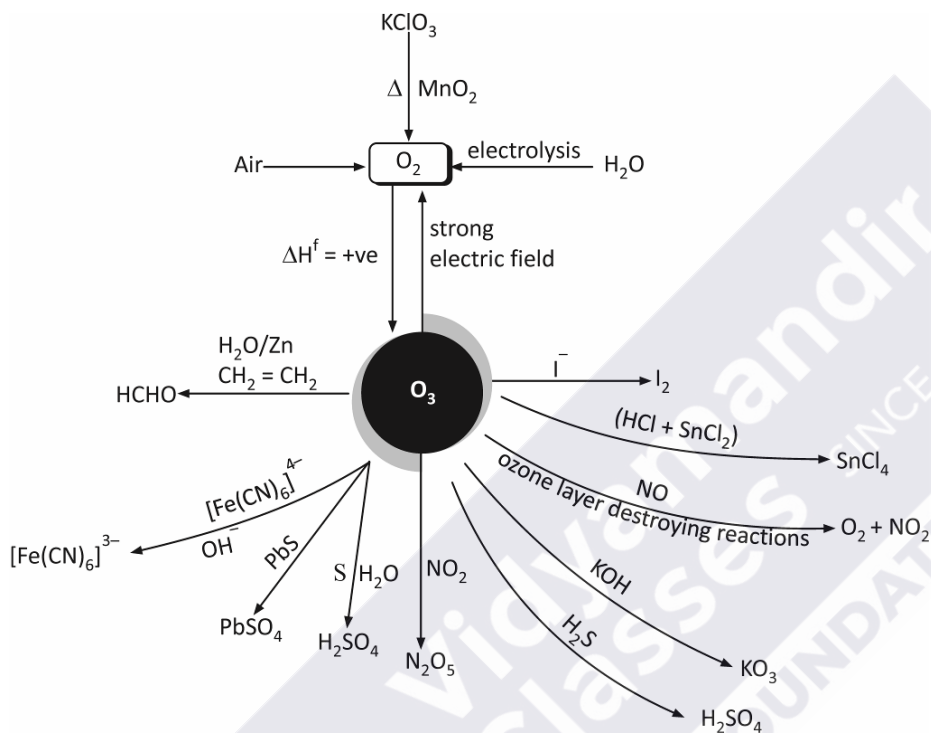
31. **Ozone**

Gaseous O_3 : Pale Blue

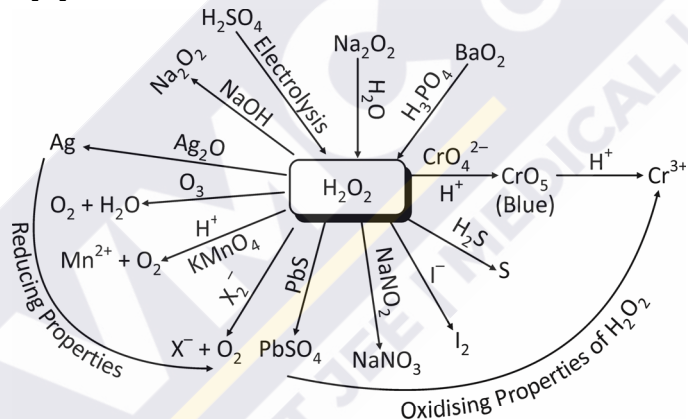
liquid O_3 : dark blue

Solid O_3 : Violet black

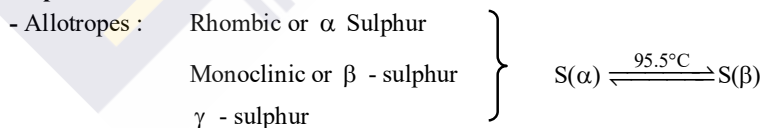
32. **Reactions Involving O_2 and O_3**



33. **H_2O_2**



34. **Sulphur**



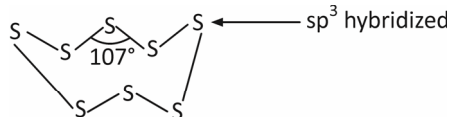
At 369 K both forms are stable. It is called transition temperature. Both of them have S_8 molecules.

The ring is puckered and has a crown shape. Another allotrope of sulphur - cyclo S_6 ring adopts a chair form.

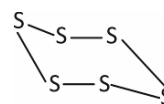
S_2 is formed at high temperature (~ 1000 K). It is paramagnetic because of 2 unpaired electrons present in anti bonding $2p^*$ orbital like O_2 .

- Amorphous forms : Plastic sulphur
Colloidal sulphur

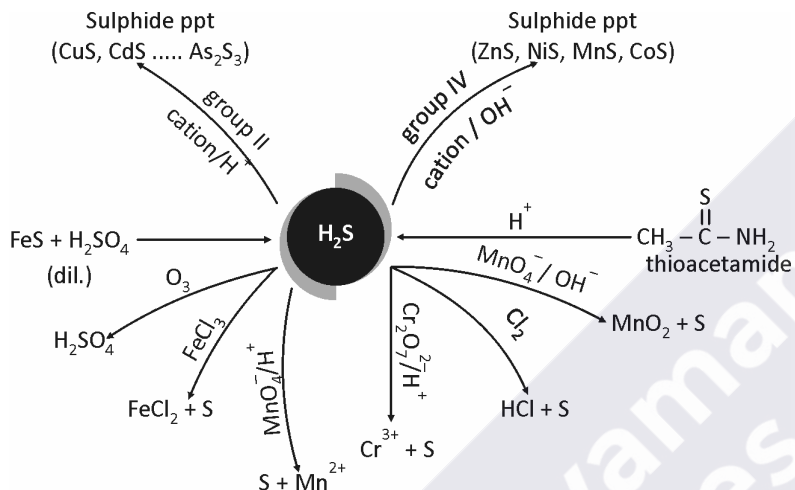
Exist as S_8 (most stable)



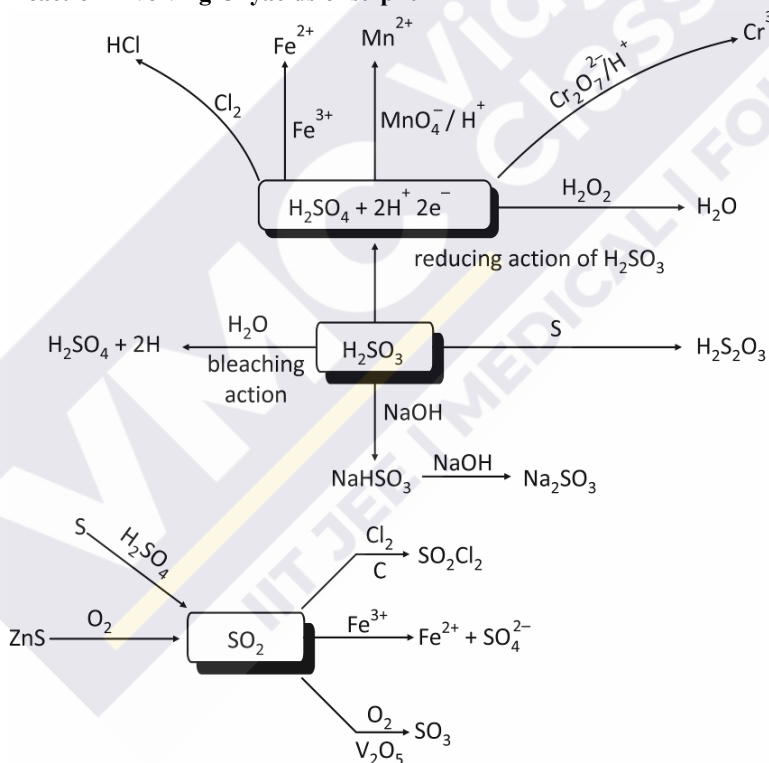
(S_6 form)



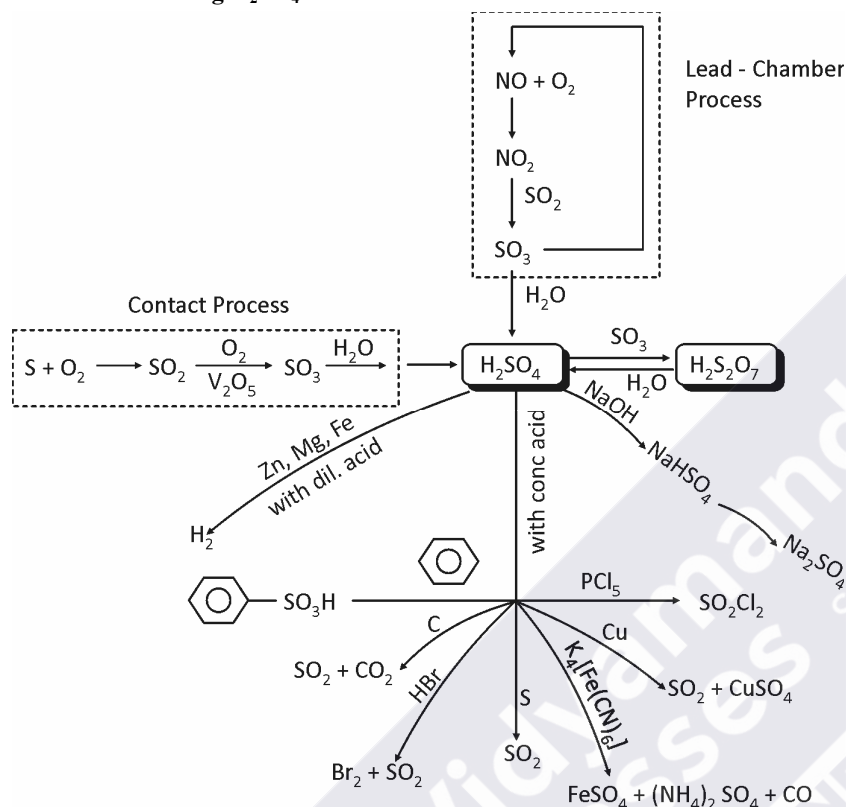
35. Reactions involving H_2S



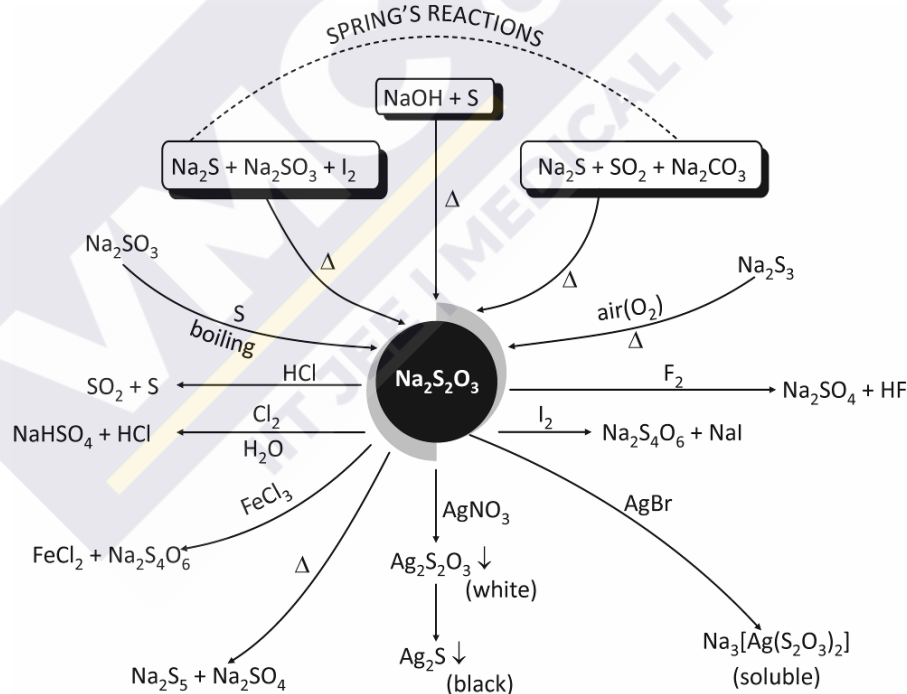
36. Reaction Involving Oxyacids of sulphur



37. Reactions Involving H_2SO_4



38. Reactions Involving $\text{Na}_2\text{S}_2\text{O}_3$ (Sodium Thiosulphate)



39. Reactions Involving Halogens

Reaction	Comment
With H₂O $2F_2 + 2H_2O \longrightarrow 4H^+ + 4F^- + O_2$ $2X_2 + 2H_2O \longrightarrow 4H^+ + 4X^- + O_2$ $X_2 + H_2O \xrightarrow{\text{disproportionate}} X^- + HOX + H^+$ With H₂ $H_2 + X_2 \longrightarrow 2HX$ With metals $2M + nX_2 \longrightarrow 2MX_n$ With CO $CO + X_2 \longrightarrow COX_2$ With P $2P + 3X_2 \longrightarrow 2PX_3$ $2P + 5X_2 \longrightarrow 2PX_5$ With S $2S + X_2 \longrightarrow S_2X_2$ $S + 2Cl_2 \longrightarrow SCl_4$ $S + 3F_2 \longrightarrow SF_6$ With H₂S $H_2S + X_2 \longrightarrow 2HX + S$ With SO₂ $SO_2 + X_2 \longrightarrow SO_2X_2$ With NH₃ $8NH_3 + 3X_2 \longrightarrow N_2 + 6NH_4X$ With halogens $X_2 + X'_2 \longrightarrow 2XX'$ With cold NaOH $2NaOH + 2F_2 \longrightarrow 2NaF + OF_2 + H_2O$ $2NaOH + X_2 \longrightarrow NaX + NaOX + H_2O$ Cl_2 reacts with dry slaked lime forming bleaching powder $Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$ With hot NaOH $6NaOH + 3X_2 \longrightarrow 5NaX + NaXO_3 + 3H_2O$	Vigorous reaction with F₂ ; atmospheric O₂ can oxidise I⁻ to I₂ hence, reverse reaction Cl₂ > Br₂ > I₂ (F₂ does not disproportionate). All the halogens with Br₂ photochemical reaction, with I₂ very slow even at high temperature. Most metals from halides. Only Cl₂, Br₂ from carbonyl halide For F, Cl, Br, I For F, Cl, Br Cl, Br All the halogens oxidise H₂S(S²⁻) to S F and Cl F, Cl, Br Interhalogen compounds (X = Cl, Br) I₂ does not react under the same condition. (X = Cl, Br, I)

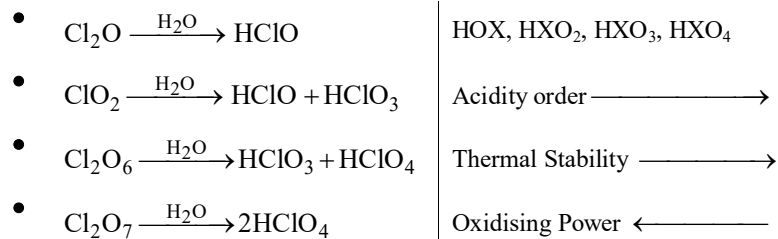
40. Halogens

Reaction with	F ₂	Cl ₂	Br ₂	I ₂
Preparation of Gas	Light - yellow gas	Yellow green gas	Brown liquid	Violet solid
	most reactive element known, strong O.A.	toxic, has pungent suffocating odour		Crystallizes as black flakes has slight lustre.
	Moisson was awarded Nobel Prize for his methods of preparing F₂ : CaF₂ + conc. H₂SO₄ → HF HF + KF → K⁺[HF₂]⁻ K⁺[HF₂]⁻ + KF $\xrightarrow{\text{electrolysis}}$ F₂ + K₂	□ Lab : MnO₂ + 4HCl → MnCl₂ + 2H₂O + Cl₂ □ Deacon's Process : It is then dried with CaO, P₄O₁₀ or H₂SO₄ HCl + O₂ $\xrightarrow{\text{CuCl}_2}$ Cl₂ + H₂O □ Commercially it is also obtained by electrolysis of NaCl(aq) in the manufacture of NaOH or elect. of NaCl(l) in the manufacture of Na	Obtained from Sea water and brine lakes by passing Cl₂ as follows : Cl₂ + 2Br⁻ → Cl⁻ + Br₂ Br₂ + Na₂CO₃ → NaBr + NaBrO₃ $\downarrow \text{H}^+$ Br₂ $\xleftarrow{\text{CuCl}_2}$ HBr + HBrO₃ pure	Chile saltpetre is mainly NaNO₃ but contains traces of NaIO₃. NaNO₃ is crystallised out and left NaIO₃ is divided into two parts NaIO₃ $\xrightarrow{+\text{NaHSO}_3}$ NaIO₃/H⁺ + SO₄²⁻ $\downarrow$ $\leftarrow \dots \dots \dots$ I₂ (solid) Purification is done by sublimation.
	H₂O F₂ + H₂O → 2H⁺ + 2F⁻ + $\frac{1}{2}$ O₂ (OA) (spontaneous and strongly exothermic)	Cl₂ + H₂O $\xrightleftharpoons[\text{K}^+]{\text{Cold OH}^-}$ ClO⁻ + Cl⁻ Cl₂ + H₂O $\xrightarrow{\text{hot OH}^-}$ ClO₃⁻ + Cl⁻	Br₂ + H₂O $\xrightarrow{\text{OH}^-}$ Br⁻ + BrO⁻	Sparsingly soluble in water but dissolves if I⁻ are present (as soluble I₃⁻ forms) I⁻ + I₂ → I₃⁻
NH ₃	X ₂ reacts with NH ₃ to give NH ₂ and NH ₄ X (X = F, Cl, Br)			
Others	All halogens oxidise H ₂ S to S ; form SO ₂ X ₂ with SO ₂ ; trihalides with As, Sb, Bi ; penta halides (except I) with As, Sb, Bi.			
Uses	Used in making freon (CFC's)	□ Used as a bleaching agent Cl₂ + H₂O → HOCl □ and make poisonous gases like SO₂Cl₂	It is used to make flame retardant and in organic chemistry.	It is used to make tincture of iodine (I ₂ + KI) and in titrations.

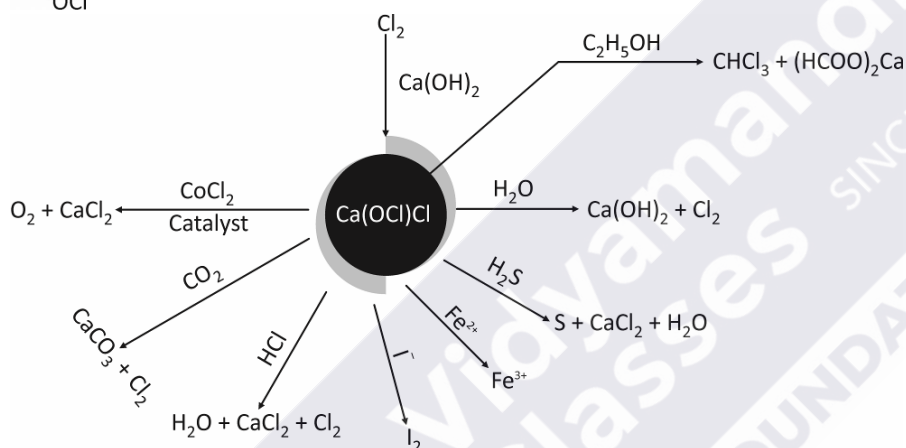
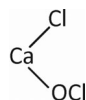
41. Oxides

a	Cl ₂ O	Cl ₂ + Na ₂ CO ₃ + H ₂ O $\xrightarrow{\Delta}$ Cl ₂ O + NaHCO ₃ + NaCl	
b	ClO ₂	AgClO ₃ + Cl ₂ → AgCl + ClO ₂ + O ₂	
c	Cl ₂ O ₇	HClO ₄ $\xrightleftharpoons{\text{P}_4\text{O}_{10}}$ Cl ₂ O ₇ + H ₂ O	
D	Cl ₂ O ₆	ClO ₂ + O ₃ → ClO ₃ $\xrightarrow{\text{dimer}}$ Cl ₂ O ₆ → [ClO ₂] ⁺ [ClO ₄] ⁻ (solid state)	or

42. Oxyacids of Cl₂



43. Bleaching Powder :



44. Xenon Fluoride

Reactions Involving Xenon Fluorides :-

